

KARTHIK KUNNAMKUMARATH

AI Engineer · Generative AI & RAG · AI-Assisted Development · Python & Full-Stack · PMP® · CSM®

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SUMMARY

AI Engineer shipping generative-AI and automation systems in production, working AI-first — using AI coding assistants to write, refactor, and extend Python and JavaScript, then reviewing generated output for correctness, security, and quality before it ships. Currently leading AI workflow transformation across four commonly owned companies at IBX Services, and building an enterprise GenAI document-generation platform at Kinectrics on FastAPI + GPT-4-class LLMs with configurable RAG. Strong record of translating business requirements into buildable specifications — prior Python automation and systems-integration work at Estée Lauder cut issue resolution 30% and operational downtime 40%. PMP® and CSM® certified; postgraduate certificate in Applied AI Solutions Development.

TECHNICAL SKILLS

AI-Assisted Development Claude Code · Cursor · GitHub Copilot · Claude · ChatGPT · prompt engineering and iterative refinement · prompt/pattern documentation · reviewing AI-generated code for security, correctness, and quality · AI-first delivery workflows

Generative AI & ML Python · GPT-4-class LLMs · OpenAI API · LangChain · LangGraph · RAG pipelines (vector, BM25, hybrid, index-free) · vector embeddings · FAISS · sentence-transformers · Hugging Face · PyTorch · TensorFlow · NLP · computer vision · conversational AI

Full-Stack & Cloud FastAPI · REST APIs · webhooks and event-driven integrations · JavaScript (ES6+) · HTML5 / CSS3 / Bootstrap · JSON schema validation · SQL · Docker · Google Cloud Vertex AI · Azure OpenAI · AWS

Engineering & Delivery Git / GitHub · code review · documentation · debugging · data structures · AI-assisted test generation · CI/CD fundamentals · software security practices · API feasibility assessment · Agile / Scrum · Jira · Confluence · requirements gathering · stakeholder management

EXPERIENCE

IT & AI Specialist · IBX Services Inc.

Apr 2026 – Present

North York, Toronto, ON · Sole IT/AI lead across four commonly owned companies (IBX, Tiron, Unica, RBR)

- Leading **AI workflow transformation** across four operating companies — designing a field-capture-to-structured-record pipeline that turns technician phone submissions, photos, and voice notes into structured job records, replacing manual re-keying across restoration and construction operations.
- Ran **API feasibility assessment** across the core operational stack before building: confirmed CompanyCam exposes a full REST API with webhooks (selected as integration backbone) and established Restoration Manager has no public write API — designing a human-verified handoff rather than an unsupported integration.
- Translating operational requirements into buildable specifications — conducting structured workflow-discovery interviews with staff across all four companies and maintaining a **Living Business Map** of systems, data flows, and handoffs that drives the AI automation roadmap and prioritisation.
- Prototyping document-to-record automation: **Xactimate PDF** → **QuickBooks invoice drafts** and voice/photo → emergency-sheet drafts, with human review gates on every generated output before it enters a system of record.
- Shipped internal Python tooling and hardened infrastructure — a **Python/Pillow badge-generation pipeline** producing 29 print-calibrated employee IDs, Hikvision access provisioning, full data recovery after a NAS RAID failure, and standardised credential management on Bitwarden.

AI Solutions Developer · Kinectrics

Feb 2026 – Sep 2026

Toronto, ON (Remote) · AI Document Generator — enterprise GenAI platform, 6-engineer team

- Owned design and implementation of the **Stateful Pipeline Architecture** — a one-click preset system bundling JSON templates, RAG parameters, and Base64-encoded DOCX/PDF assets into portable .pipeline.json files, with a runtime engine that decodes assets, injects parameters, executes generation, and cleans up temporary files.
- Integrated the pipeline layer across **FastAPI backend and web front end** — implementing JavaScript state-management guardrails that lock manual inputs once a pipeline is selected, preventing users from overriding standardised enterprise configurations.
- Built with **AI coding assistants** (Claude, ChatGPT) to write, refactor, and extend Python and JavaScript — iterating on prompts to reach working implementations and reviewing all generated code for correctness, security, and standards alignment before merge.
- Contributed to prompt and schema design across the platform — section-level JSON schemas specifying objective, tone, length, and format — and to a configurable **RAG layer supporting four retrieval strategies** (vector, BM25, hybrid, index-free); reduced report-drafting time from hours to minutes.

Systems Analyst / Maintenance Project Supervisor · Estée Lauder

Jun 2022 – Apr 2026

Markham, ON · Enterprise IT integration, automation, and analytics

- Translated operational stakeholder requirements into a **Python-based workflow-automation layer** integrating Jira and Maximo (CMMS) — cut issue resolution time **30%** and eliminated 20+ hours per week of manual reporting.
- Delivered executive-facing **Power BI dashboards** translating operational data into business-ready KPIs, and tracked production trends to iterate on system improvements; led IT integrations that reduced operational downtime **40%**.

Project Manager, IT & Construction · Sangeeth Builders

Jan 2017 – Jan 2019

Kerala, India · Delivered \$500K+ in IT and construction programs at 100% on-time completion; implemented a digital inventory-management system with full requirements documentation and user training, cutting procurement and reporting delays 50%.

SELECTED PROJECTS

Offline RAG Chatbot for Regulatory Document Search

GenAI · Conversational AI

LangChain · FAISS · Llama 3 (Ollama) · Streamlit · Python

- Engineered a **100% offline Retrieval-Augmented Generation chatbot** for secure Q&A over technical manuals with no external API exposure — PDF ingestion, semantic chunking (1000/200 overlap), all-MiniLM-L6-v2 embeddings persisted to FAISS, k=4 similarity retrieval, and locally hosted Llama 3 at temperature 0 for deterministic answers.
- Built page-level source-citation rendering from document metadata so every generated answer is traceable to source — a practical guardrail for evaluating LLM output quality in compliance-sensitive workflows.

RoadSense AI · End-to-End Computer Vision Application

AI/ML · Full-Stack

YOLO11m · Faster R-CNN · Gradio · Folium · Python

- Benchmarked two detection architectures on RDD2022 (7,634 images, 4 classes) — **YOLO11m mAP50 0.491** versus Faster R-CNN 0.414 — and deployed the winning model behind a Gradio interface with severity scoring, repair-cost estimation, and geospatial dispatch routing.

EDUCATION & CERTIFICATIONS

Postgraduate Certificate, Applied AI Solutions Development · George Brown College 2025 – 2026

Postgraduate Certificates, Project Mgmt & Process Quality Eng. · Conestoga College 2019 – 2020

B.Tech, Aerospace Engineering · Amrita School of Engineering 2013 – 2017

Certifications PMP® (PMI, valid 2029) · CSM® (Scrum Alliance, valid 2026)